Water controls the fields of metastable olivine in cold subducting slabs

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[1] We have measured the rate of the olivine-wadsleyite transformation in Mg_2SiO_4 under hydrous condition by in-situ X-ray diffraction method. The concentration of hydroxyl of samples was estimated using infrared spectroscopy. Combined with previous kinetic data on this transformation, a total of 15 growth rate data obtained at 13.4–15.8 GPa, 1003–1373 K, and 660–5000 ppm weight H_2O have been used to establish the pressure, temperature and OH content dependences. Results demonstrate that the growth rate in the olivine-wadsleyite transformation is proportional to OH content to the power of about 3. The depth of the olivine transformation significantly changes with the water content in cold slabs, which must be considered to estimate fields of the metastable olivine.
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1. Introduction

[2] It has been proposed that metastable olivine possibly exists beyond the equilibrium boundary in the interior of cold slabs because the olivine-wadsleyite (ringwoodite) transformation is sluggish due to low temperatures [e.g., *Sung and Burns*, 1976; *Rubie and Ross*, 1994]. Presence of the metastable olivine has important implications for dynamics of subducting slabs, such as buoyancy force against subduction, changes of stress state in slabs [e.g., *Yoshioka et al.*, 1997], and origin of deep earthquakes [e.g., *Green and Burnley*, 1989; *Kirby et al.*, 1996]. Recent studies on modeling the fields of metastable olivine in the slabs [*Mosenfelder et al.*, 2001; *Marton et al.*, 2005] have revealed that the depth of the olivine transformation is shallower than the depths of the deepest seismicity, which is against the transformational-faulting model for the origin of deep earthquakes.

[3] Olivine and wadsleyite can dissolve hydrogen in their crystal structures as point defects. It is known that such hydrogen enhances the diffusivity in both olivine [*Hier-Majumder et al.*, 2005] and wadsleyite [*Kubo et al.*, 2004a], and the rate of the olivine-wadsleyite transformation [*Kubo et al.*, 1998]. However, experimental kinetic data, in which the OH content of samples is not known, have been used to estimate the field of metastable olivine so far. Recently, *Kubo et al.* [2004b] have reported experimental results on growth kinetics in the olivine-wadsleyite transformation with the OH content of samples. Although the experiments were conducted under nominally anhydrous conditions, some amounts of hydroxyl were apparently contained in the samples.

[4] In this study, we have measured the olivine-wadsleyite transformation rate under deliberately hydrous conditions in wider ranges of OH contents by in-situ X-ray diffraction method. OH contents of samples have been determined by infrared spectroscopy. Some portion of kinetic data of the present study has been reported preliminarily by *Ohtani et al.* [2004] without measuring OH contents. We have reexamined the data carefully including the OH content and added new experimental data. Combined with previous growth rate data with OH content [*Kubo et al.*, 2004b], we have determined the pressure, temperature, and OH content dependences of the growth rate in the olivine-wadsleyite transformation. Based on these experimental results, we discuss effects of water on the depth of the olivine transformation in cold slabs.

2. Experimental Method

[5] High-pressure in-situ X-ray diffraction experiments have been carried out using a Kawai-type high-pressure apparatus "SPEED-1500" installed in beamline BL04B1 at SPring-8, Japan. Details of high-pressure experiments are given in elsewhere [*Kubo et al.*, 2004b]. The sample assembly used in this study is as same as that by *Kubo et al.* [2004b] except for the sample capsule. In the present study, Ag-Pd alloy (inner and outer diameters are 1.25 and 1.40 mm, respectively) was used as the sample capsule to minimize fluctuations of the OH content during experiments. Two kinds of starting material were used in this study, Mg_2SiO_4 + 2000 wt.ppm H_2O and Mg_2SiO_4 + 5000 wt.ppm H_2O. These were prepared as mixtures of MgO, SiO_2, and $Mg(OH)_2$ powders.

[6] The powder mixture of the starting material was first annealed to synthesize polycrystalline olivine under hydrous condition. The annealing condition was 10 GPa and 1273 K for the system of Mg_2SiO_4 + 5000 wt.ppm H_2O, and 1373 K for Mg_2SiO_4 + 2000 wt.ppm H_2O. The heating duration was 100 minutes in both cases. In addition to diffraction peaks of olivine, small amounts of MgO and high-clinoenstatite $MgSiO_3$ were identified by in-situ X-ray observations. It is thought that these minor phases did not affect the transformation rate significantly because these were still present after the transformation. Two samples with varying OH contents were quenched after the annealing prior to transformation in order to measure grain size of parental olivine. The grain size was estimated to be 5 ± 3 and 7.5 ± 2 microns in the system of Mg_2SiO_4 + 2000 wt.ppm H_2O and Mg_2SiO_4 + 5000 wt.ppm H_2O, respectively, from SEM observations.

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Table 1. Experimental Conditions, OH Contents and Estimated Values of Growth Rates[a]

Run No.	Temperature, K	Pressure, GPa	Overpressure, GPa	Duration,[b] min	OH Contents, wt ppm H_2O	Growth Rate, ms^{-1}
			Mg_2SiO_4 + 5000 wt. ppm H_2O			
S824	1173	15.2	2.7	26	3884 (388)	7.0 (4.9) × 10^{-9}
S826	1083	14.2	2.0	18	5000 (622)	2.4 (1.6) × 10^{-9}
S918[c]	1003	14.5	2.6	590	2063 (741)	3.1 (2.0) × 10^{-11}
			Mg_2SiO_4 + 2000 wt. ppm H_2O			
S1031	1173	15.1	2.6	28	3626 (615)	4.5 (1.5) × 10^{-9}
S1029[d]	1103	14.3	2.0	82	2176 (677)	6.7 (2.5) × 10^{-10}
S961	1093	14.3	2.0	78	2907 (487)	6.5 (2.1) × 10^{-10}

[a]Numbers in parentheses represented the uncertainties.
[b]Overall heating duration at the desired pressure and temperature conditions during the transformation step.
[c]Failure in the seal of water due to a crack found in the capsule recovered.
[d]Temperature was estimated from the power supply. The temperature uncertainty for this run is estimated to be ±50 K.

[7] After annealing, the sample was compressed to the stability field of wadsleyite, and then heated to the target temperature. Kinetics of the olivine-wadsleyite transformation was observed by time-resolved X-ray diffraction measurements every 30–300 seconds with energy-dispersive method using a solid-state detector. Temperature was measured by a W3%Re–W25%Re thermocouple and pressure was evaluated from equation of state of NaCl [*Decker*, 1971]. The thermocouple junction and the pressure marker were placed next to the sample capsule, so that we can take diffraction patterns of the sample and the pressure marker independently very close to the thermocouple. The overpressure was estimated based on the equilibrium boundary determined by *Morishima et al.* [1994]. In this way, we observed the olivine-wadsleyite transformation kinetics at six different points of 14.2–15.2 GPa and 1003–1173 K under hydrous condition.

[8] Unpolarized infrared absorption spectra of the polycrystalline sample recovered were obtained in air using Fourier-transform infrared spectrometers (JASCO MFT-2000 at Tohoku University, and JASCO FT-IR700 combined with Micro-20 at the Center of Advanced Instrumental Analysis, Kyushu University). Details of FTIR measurements are described in elsewhere [*Kubo et al.*, 2004b]. OH contents of the sample were calculated using the calibration by *Paterson* [1982]. It has been reported that unpolarized IR measurements with the Paterson's calibration is inaccurate by a factor of 3 to 4 for some minerals [e.g., *Bell et al.*, 2003]. FTIR measurements in the present study also possibly have these uncertainties.

3. Results and Discussion

[9] The experimental conditions and results are summarized in Table 1. We have successfully obtained time-resolved X-ray diffraction profiles from the sample sealed in the Ag-Pd capsule. We estimated transformed volume fraction assuming that the relative integrated intensities of the diffraction peak are proportional to the weight fraction of the phase. The diffraction lines 031, 211, 222 + 240, 061 + 133, 004 + 251 of olivine and 310, 112, 400, 042, 442, 080 in wadsleyite were used for this purpose.

[10] In the olivine-wadsleyite transformation, the grain-boundary reaction is the dominant mechanism in both nominally dry and wet conditions [e.g., *Kubo et al.*, 1998; *Kubo et al.*, 2004b]. Fitting of the obtained kinetic data[1] by the Avrami rate equation yields the *n*-value of nearly 1 in all of runs. This indicates that the saturation of nucleation site occurs at the initial stage of the transformation [*Cahn*, 1956]. This can be explained by high nucleation rates under relatively large overpressure conditions (2–3 GPa) of the present study. In this case, the overall transformation rate is controlled only by the growth. Therefore we estimate the growth rate $\dot{x}$ in the transformation by fitting the following rate equation to kinetic data obtained.

$$V = 1 - \exp(-2S\dot{x}t) \quad (1)$$

where V is the transformed volume fraction, S is area of grain-boundary of parent phase (=3.35/d), d is the grain size of the parental olivine, and t is time [*Cahn*, 1956]. Estimated values of the growth rate are listed in Table 1.

[11] Figure 1 shows unpolarized infrared spectra of polycrystalline samples recovered after the transformation. Strong absorption bands were observed in 3408, 3350, and 3322 cm^{-1}, which can be assigned to the O-H vibration in wadsleyite [e.g., *Kohn et al.*, 2002]. Infrared spectra obtained from the polycrystalline sample of the present study generally show a broad absorption band from 2800 to 3600 cm^{-1}, which is consistent with the study of *Keppler and Rauch* [2000]. The broad absorption band may be due to the presence of hydrous species on grain boundaries. We estimated OH contents of the bulk sample both from sharp and broad absorption bands. The average OH content from 3–8 points in each sample is listed in Table 1. The OH content of the sample in the system of Mg_2SiO_4 + 5000 wt.ppm H_2O recovered after the annealing prior to the transformation was also measured and estimated to be 3906 ± 547 ppm weight H_2O. The variation in the OH content even in the same starting material may indicate that the OH content of samples varied throughout the experiment.

[12] Combined with the results of the present study and *Kubo et al.* [2004b], a total of 15 growth rate data obtained at 13.4–15.8 GPa, 1003–1373 K, and 660–5000 ppm weight H_2O are used to construct the pressure, temperature and OH content dependences on the growth kinetics in the olivine-wadsleyite transformation of Mg_2SiO_4. The growth

[1]Auxiliary material is available at ftp://ftp.agu.org/apend/gl/2005GL023398.

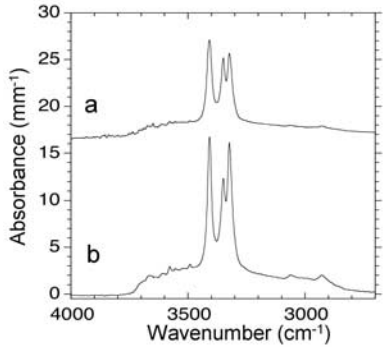


Figure 1. Unpolarized infrared spectra of samples recovered after the transformation. (a) Mg_2SiO_4 + 2000 wt.ppm H_2O (Run S1029), (b) Mg_2SiO_4 + 5000 wt.ppm H_2O (Run S824). Small absorption peaks near 2900 cm^{-1} may be derived from the acetone treatment during sample preparation [*Kubo et al.*, 2004b; *Fumagalli et al.*, 2001].

in a polymorphic transition is controlled by interface kinetics. The growth rate law of *Turnbull* [1956], which is a theoretical equation derived based on atomistic considerations of diffusion across the transforming interface, has been modified to an empirical equation by inserting a simple power-law dependence on OH content as follows,

$$\dot{x} = ATC_{OH}^{n} \exp\left(-\frac{\Delta H_a + PV^*}{RT}\right)\left[1 - \exp\left(\frac{-\Delta G_r}{RT}\right)\right] \quad (2)$$

where A is a pre-exponential factor, T is absolute temperature, C_{OH} is the concentration of hydroxyl, n is the OH content exponent, ΔH_a is the activation enthalpy for growth, P is pressure, V^* is the activation volume for growth, R is the gas constant, and ΔG_r is the free energy change of the transformation. This rate equation was fitted to the growth rate data by a weighted least-squares procedure, which yields ln A = -18.0 ± 3.8 ms^{-1} $wt.ppmH_2O^{-3.2}$, n = 3.2 ± 0.6, ΔH_a = 274 ± 87 kJ/mol, and V^* = $(3.3 \pm 3.8) \times 10^{-6}$ m^3/mol. Results of the fitting are shown in Figure 2 as plots of the pressure, temperature, and OH content against the growth rate.

[13] Recent studies have revealed that the strain rate [e.g., *Mei and Kohlstedt*, 2000] and Fe-Mg interdiffusion rate [*Hier-Majumder et al.*, 2005] of olivine increase almost linearly with increasing water fugacity, which can be understood based on the point defect chemistry in the crystal. On the other hand, water also enhances the grain-boundary processes such as grain growth and dynamic recrystallization in both olivine and wadsleyite [e.g., *Young et al.*, 1993; *Jung and Karato*, 2001]. It has been suggested that the enhancement of grain-boundary mobility by water is more significant than that of intragranular dislocation recovery [*Jung and Karato*, 2001].

[14] The rate of interface-controlled growth depends on the mobility of inter-phase boundary and the thermodynamic driving force. The OH content exponent of about 3 for the growth rate in the olivine-wadsleyite transformation is larger than that for the creep rate and the diffusivity. Because water does not affect the thermodynamic driving force significantly, our results suggest that water has a large effect on the mobility of inter-phase boundary. Presence of hydrous species on grain boundary, which was inferred from the broad absorption band in infrared spectra, possibly has an important role for the enhancement of the mobility. However, the interpretation of the broad absorption band is problematic at the present stage. It is not clear that water was stably present along grain boundaries in situ at high pressure and temperature conditions because of the high solubility of hydrogen in wadsleyite. Chemical states of OH species in the sample may change upon quenching and decompression. In situ infrared measurements are needed to assess these issues.

[15] This study conclusively demonstrates that water greatly enhances the growth rate during the olivine-wadsleyite transformation. Because the depth of the olivine transformation in cold slabs is controlled by the growth kinetics [*Rubie and Ross*, 1994], effects of water on the growth kinetics must be considered to estimate fields of metastable olivine. Although there are limitations when applying the currents results to the Earth because we have studied Mg end member system, we have made a simple calculation to show changes of the depth of the transformation with the OH content in a cold slab based on the growth kinetics determined in this study (Figure 3). We assumed a grain-boundary reaction controlled by growth kinetics, similar growth kinetics between wadsleyite and ringwoodite, and an olivine grain size of 5 mm. The olivine-wadsleyite phase boundary determined by *Morishima et al.*

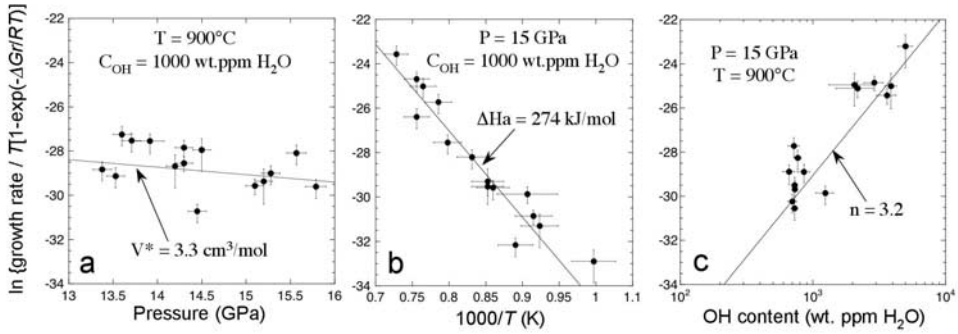


Figure 2. (a) Pressure, (b) temperature, and (c) OH content dependences of the growth rate obtained in the present study and *Kubo et al.* [2004b]. For plotting purposes, the growth rate data are plotted after corrections to given P, T, and OH content conditions using kinetic parameters determined in this study.

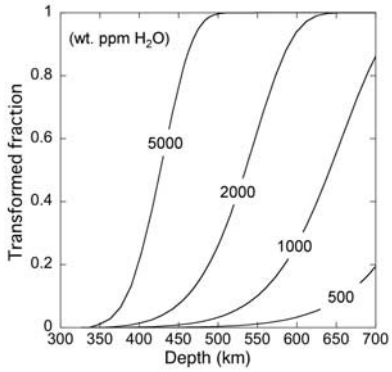


Figure 3. Depths of the olivine transformation with water contents in the cold slab. The temperature at 660 km of 873 K, the thermal gradient of 0.6 K/km, and the vertical subduction speed of 12 cm/year are used as the pressure-temperature-time path of the cold slab.

[1994] was used as an equilibrium boundary in the calculation. The following equation was used to estimate extent of transformation in the cold slab with a vertical subduction speed of 12 cm/year,

$$V = 1 - \exp\left[-2S \int_0^t \dot{x}(z)dz\right] \quad (3)$$

where $\dot{x}(z)$ is the growth rate calculated at each time step z [*Mosenfelder et al.*, 2001]. The temperature at 660 km depth of 873 K and the thermal gradient of 0.6 K/km are used as a representative coldest slab geotherm, which is similar to Slab D model by *Kirby et al.* [1996].

[16] Figure 3 clearly shows that the depth of the olivine transformation drastically changes with the water content in the cold slab although effects of latent heat release [*Mosenfelder et al.*, 2001], more realistic thermal model including variable thermal conductivity [*Marton et al.*, 2005], and effects of intracrystalline transformation [*Kerschhofer et al.*, 2000] should be incorporated in the calculation in the future. According to our preliminary calculations in Figure 3, the metastable olivine can survive greater than the 650 km depths in cold slabs containing the water of less than 500 wt.ppm H_2O, which is identical to the depths of the deepest seismicity. Whereas in cold slabs containing the water of 5000 wt.ppm H_2O, the depth of the transformation reduces to about 450 km depths even in the cold slab. Thus, water would control the field of the metastable olivine and dynamics of subducting cold slabs.

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